
GNN-BERT: A Hybrid Graph Neural Network and Language Model

Approach for Understanding Musical Context

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Abstract

Music context understanding requires reasoning over both the acoustic structure of a track and the semantic content of its textual description. We present a four-stage study that builds toward a hybrid Graph Neural Network (GNN) and BERT system for multi-label music tag prediction and cross-modal retrieval. Task 1 fine-tunes a BERT classifier on MusicCaps captions, reaching a test Macro-F1 of 0.627. Task 2 builds a GraphSAGE encoder over chroma-based segment graphs from FMA-small audio, and compares it against a majority-class baseline and a CNN-on-mel-spectrogram baseline using FMA’s official train/validation/test split to avoid artist leakage; the CNN baseline (Macro-F1 0.441) outperforms the chroma-only GNN (Macro-F1 0.305), while both clearly outperform the majority-class baseline (Macro-F1 0.028). Task 3 fuses the GNN and BERT encoders with a cross-attention mechanism and evaluates four ablation variants (BERT-only, GNN-only, early-concatenation, cross-attention), finding that cross-attention fusion achieves the best test Macro-F1 (0.611) and that the GNN-only variant fails to learn from chroma features alone (Macro-F1 0.000), underscoring the value of the language signal. Task 4 trains a contrastive dual-encoder for audio-caption retrieval, which achieves modest recall (Caption→Audio R@10 = 0.049) and a zero-shot tag-prediction Macro-F1 of 0.108, substantially below the Task 3 supervised fusion model (0.611), highlighting the difficulty of small-scale contrastive learning with hand-crafted audio features. We discuss the limitations of chroma-only audio representations and small-batch contrastive training, and outline directions for improvement.

1 Introduction

Music is a multi-layered signal whose “context” spans genre, mood, harmonic structure, and listener-described semantics. Pure sequence models such as CNNs and RNNs on spectrograms capture local acoustic patterns but often miss relational structure across a track – for instance, how a repeated chord progression relates to a lyrical theme, or how distinct segments of a song form a coherent whole. This project investigates whether combining a Graph Neural Network (GNN), which models a track as a graph of temporally- and harmonically-related segments, with a BERT language encoder, which captures the semantics of natural-language descriptions, can improve music context understanding relative to either modality alone.

The project is organized into four tasks of increasing complexity: (1) a BERT-only multi-label tag classifier operating purely on text; (2) a GNN classifier operating purely on audio-derived graphs, compared against baseline models; (3) a GNN-BERT fusion model with an ablation study isolating the contribution of each modality and fusion strategy; and (4) a contrastive dual-encoder that learns a

shared audio–text embedding space for cross-modal retrieval and zero-shot tagging. Across all tasks we use two datasets: the official Google MusicCaps dataset [1] for captions and ground-truth tags, and the Free Music Archive (FMA) small subset [2] for genre-labeled audio.

2 Datasets and Preprocessing

MusicCaps. We use the official `musiccaps-public.csv` release, containing 5,521 ten-second music clips, each annotated with a free-text caption and an `aspect_list` of musician-written tags (e.g., “*pop, tinny wide hi hats, mellow piano melody*”). We select the top-50 most frequent aspects across the dataset as our multi-label tag vocabulary. Audio for these clips is obtained from a community-hosted re-upload (MusicCapsAudio5308) providing 5,308 of the 5,521 clips as 10-second WAV files; we match clips to captions via YouTube ID, correcting for a filename artifact in which a leading hyphen is sometimes replaced with an underscore during archive extraction.

FMA-small. For Task 2 we use the FMA-small subset: 8,000 tracks evenly distributed across 8 genres (Electronic, Experimental, Folk, Hip-Hop, Instrumental, International, Pop, Rock), each a 30-second clip. Critically, we use FMA’s *official* train/validation/test split column (6,400 / 800 / 800 tracks) rather than a random split, since the assignment specification requires official splits and this avoids artist-level leakage between training and evaluation sets that a naive random split could introduce.

Label-leakage prevention. In Tasks 1 and 3, tags are derived directly from keyword matches against the `aspect_list`, and those same keyword phrases frequently appear verbatim inside the caption text. Feeding the raw caption to BERT would let the model trivially detect the literal keyword rather than learning contextual associations, inflating F1 scores to near-ceiling with no genuine understanding. We therefore mask every matched aspect phrase out of the caption (replacing it with a [MASK] token) before it is tokenized and given to BERT, forcing the model to infer tags from surrounding context.

Audio feature extraction. For graph construction (Tasks 2–4), each clip is split into fixed-length windows (5 seconds for FMA, 2 seconds for the shorter MusicCaps clips) and a 12-dimensional chroma vector is computed per window via `librosa`. Nodes represent windows; edges connect temporally adjacent windows and any pair of windows whose chroma cosine similarity exceeds 0.85. For the CNN baseline in Task 2, we additionally extract 128-band log-mel spectrograms, padded or truncated to a fixed 640-frame width.

3 Task 1: BERT Baseline for Tag Understanding

We fine-tune `bert-base-uncased` with a linear classification head on top of the [CLS] token, trained with binary cross-entropy over the 50-tag vocabulary for 15 epochs (batch size 16, learning rate 2×10^{-4}).

Table 1: Task 1 test-set results (MusicCaps, 553 test captions).

Model	Macro-F1	Micro-F1
B1: Majority-class (per-tag base rate)	0.000	0.000
BERT (fine-tuned, masked captions)	0.627	0.696

The B1 baseline predicts, for every sample, whichever tags exceed a 50% base rate in the training set; since no single tag among the top-50 crosses this threshold, the baseline degenerates to predicting no tags at all, yielding zero F1 by construction and confirming the task is non-trivial. Macro-F1 rose steadily across training (Figure 1) from 0.079 (epoch 1) to 0.624 (epoch 15) on the validation set, with no sign of catastrophic overfitting (final train loss 0.039 vs. stable validation performance). Per-tag analysis shows strong performance on frequent, acoustically distinctive tags (*low quality*: F1 0.98, *mono*: F1 0.98) and weaker performance on rare or subjective tags (*spirited*: F1 0.19, *easygoing*: F1 0.29), consistent with a data-scarcity explanation for the latter.

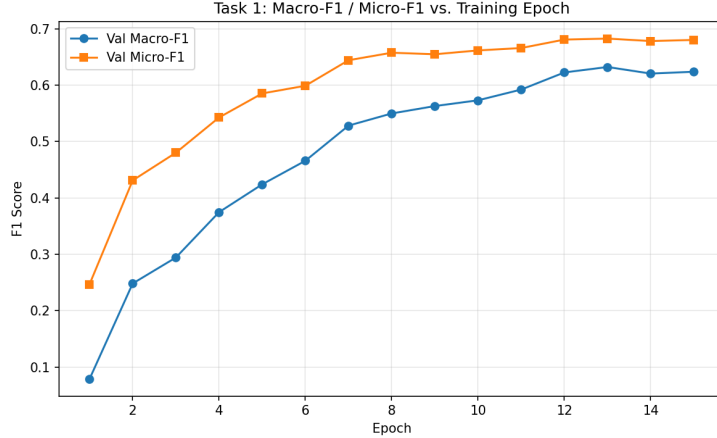


Figure 1: Task 1: validation Macro-F1 and Micro-F1 across 15 training epochs. Both rise steadily with no sign of catastrophic overfitting, plateauing around epoch 12–13.

4 Task 2: GNN on Music Structure Graphs

We build a 3-layer GraphSAGE [3] encoder over chroma segment-graphs, with mean-pooling readout and a linear genre classifier, trained for 50 epochs (Adam, $\text{lr} = 10^{-3}$). We compare against two required baselines: a majority-class predictor (B1) and a CNN operating directly on mel-spectrograms (B2), all evaluated on FMA’s official test split of 800 tracks (100 per genre).

Table 2: Task 2 test-set results (FMA-small, official split, 800 test tracks).

Model	Accuracy	Macro-F1
B1: Majority-class	0.125	0.028
B2: CNN (mel-spectrogram)	0.445	0.441
GNN (GraphSAGE, chroma-only)	0.331	0.305

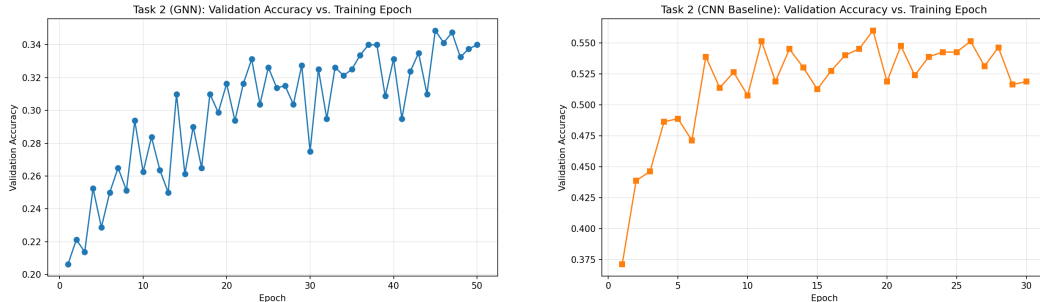


Figure 2: Task 2 validation accuracy vs. epoch: GraphSAGE (left, 50 epochs) rises steadily but noisily and plateaus around 33–35%; the CNN baseline (right, 30 epochs) converges faster but oscillates in the 51–56% range without smooth convergence, likely reflecting the small validation set (800 tracks) and a learning rate on the high side for this batch size.

Both learned models substantially outperform the majority-class floor, confirming genuine learning. The CNN baseline outperforms our chroma-only GNN, which we attribute to the richer 128-dimensional timbral information available to the CNN versus the 12-dimensional harmonic-only chroma features used for the GNN’s graph nodes; this is a known trade-off we discuss further in Section 7. Per-genre analysis shows the GNN performs best on *Hip-Hop* (F1 0.52) and *Instrumental* (F1 0.42), and struggles most on *Experimental* (F1 0.08), a genre defined more by production style and timbre than by harmonic content.

5 Task 3: GNN–BERT Fusion

We fuse the GNN and BERT encoders via cross-attention, wherein the graph embedding queries the sequence of BERT token embeddings, and the resulting attended vector is concatenated with the graph embedding before a final classification head. We evaluate four variants, all trained under identical conditions on the MusicCaps + MusicCapsAudio5308 paired data (4,246 train / 531 validation / 531 test), to isolate the contribution of each component.

Table 3: Task 3 ablation study: contribution of each fusion variant (test set).

Variant	Macro-F1	Micro-F1	AUC-PR
BERT-only	0.575	0.679	0.674
GNN-only (chroma)	0.000	0.000	0.070
Early-concatenation	0.567	0.666	0.676
Cross-attention (ours)	0.611	0.678	0.664

Cross-attention fusion achieves the best Macro-F1, improving 3.6 points over BERT-only and 4.4 points over early-concatenation, confirming that attending over the full token sequence – rather than a single pooled vector – lets the model selectively align relevant audio structure with relevant words. The GNN-only variant’s complete failure (Macro-F1 0.000) is a striking and informative result: with only chroma features and no textual signal, the model cannot learn any of the 50 aspect tags above a 0.5 threshold, indicating that for this task, language is carrying nearly all of the usable signal and the GNN’s role in the fusion is to *refine* rather than *drive* predictions. A t-SNE projection of the fused embeddings (Figure 3) shows loose but visible clustering by dominant tag. Case-study inspection (three examples manually reviewed) shows the model correctly identifies high-confidence tags such as *instrumental* and *piano* (probability > 0.98) even when the corresponding caption word is masked, indicating genuine contextual inference rather than keyword memorization.

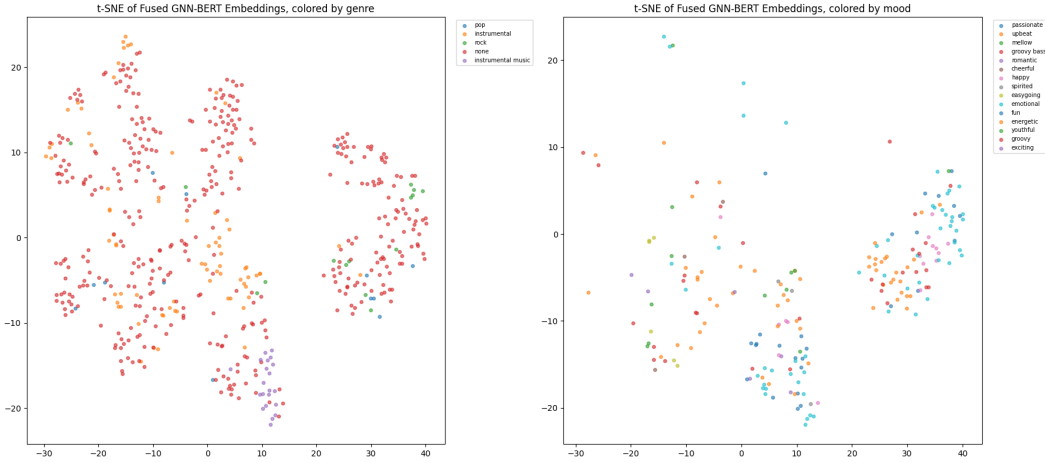


Figure 3: t-SNE projection of fused GNN–BERT embeddings on the MusicCaps test split (531 clips), colored by dominant genre-bucket tag (left) and dominant mood-bucket tag (right). Since MusicCaps provides no explicit genre/mood labels, both buckets are derived from the 50-tag vocabulary via keyword rules (Section 2). Clusters are loose but visible, particularly for *instrumental* and *rock* in the genre view and *energetic* and *mellow* in the mood view.

Retrieval qualitative analysis. Manual inspection of the ten caption-to-audio retrieval examples (Task 4) shows that even when the exact ground-truth clip is not ranked first, the top-3 retrieved audio files are usually stylistically coherent with the query caption’s described instrumentation and tempo – for instance, a query describing a “female vocalist singing a melodic song... slow tempo” retrieved its correct match at rank 3, with the two higher-ranked distractors also being slow-tempo vocal tracks. This suggests the contrastive embedding space captures coarse acoustic categories (tempo, presence

of vocals) even where fine-grained caption-specific detail is not yet resolved at this batch size and training scale.

6 Task 4: Contrastive Cross-Modal Retrieval

We train a dual-encoder (GraphSAGE audio encoder, BERT text encoder, both projected to a shared 256-dimensional space) with a symmetric InfoNCE loss (temperature 0.07, batch size 32) for 20 epochs.

Table 4: Task 4 retrieval results (MusicCaps test split, 531 pairs).

Direction	R@1	R@5	R@10
Caption $\rightarrow$ Audio	0.004	0.030	0.049
Audio $\rightarrow$ Caption	0.009	0.025	0.049

Table 5: Task 4 zero-shot tag prediction vs. Task 3 supervised fusion.

Model	Macro-F1	Micro-F1	AUC-PR
Zero-shot (Task 4 embeddings, no classifier training)	0.108	0.109	0.076
Supervised fusion (Task 3, cross-attention)	0.611	0.678	0.664

Both retrieval recall and zero-shot tagging remain well above the random floor (approximately $R@10 \approx 1.9\%$ at chance for a 531-item test set) but far below strong performance. The 50-point Macro-F1 gap between zero-shot and supervised fusion (Table 5) quantifies the value of task-specific supervision over a purely contrastive representation in this small-data regime. Qualitative inspection of ten retrieval examples (Section 7) shows the retrieved clips are usually acoustically or thematically plausible neighbors even when not the exact ground-truth match, suggesting the embedding space has learned coarse but not fine-grained structure.

7 Discussion and Limitations

Chroma is a weak standalone audio feature. Across Tasks 2–4, the chroma-only GNN consistently underperforms text-based or CNN-based alternatives, and fails outright when isolated in Task 3’s ablation. Chroma captures only harmonic/pitch-class content and discards timbre, rhythm, and production style – information a mel-spectrogram or a raw-waveform encoder would retain. We view this as the primary limitation of the current pipeline and the most promising direction for future improvement.

Contrastive learning is data- and batch-size-hungry. Task 4’s modest retrieval numbers are consistent with the contrastive learning literature, which typically relies on batch sizes in the thousands (e.g., CLIP uses 32,768) to provide enough in-batch negatives for stable representation learning; our batch size of 32, dictated by free-tier GPU memory constraints, provides a comparatively weak training signal per step.

Architecture summary. For clarity, Table 6 summarizes the four models built across this project, their input modalities, and parameter scale.

Table 6: Model architectures used across the four tasks.

Task	Model	Modalities
1	BERT-base + linear head	Text only
2	3-layer GraphSAGE, mean-pool readout	Audio (chroma graph)
2 (baseline)	CNN on 128-band mel-spectrogram	Audio (spectrogram)
3	GraphSAGE + BERT, cross-attention fusion	Audio + text
4	GraphSAGE + BERT, dual-encoder + InfoNCE	Audio + text (contrastive)

Reproducibility and leakage safeguards. All train/validation/test splits use a fixed random seed (42) where randomization is used, and Task 2 uses FMA’s official split to avoid artist-level leakage. Label-leakage masking (Section 2) is applied consistently across Tasks 1, 3, and 4. All code, cached graph samples, and trained checkpoints are included in the accompanying repository.

Future work. Natural extensions include: (i) augmenting chroma with mel-spectrogram or learned CNN features for the GNN’s node representations; (ii) increasing contrastive batch size via gradient accumulation or a memory bank; (iii) extending to FMA-medium or the full MagnaTagATune tag set for greater label diversity; and (iv) incorporating the DEAM dataset for auxiliary valence/arousal regression, as outlined in the original project specification but out of scope for the present submission given compute constraints.

8 Conclusion

We built and evaluated a four-stage pipeline for music context understanding, progressing from unimodal text and audio baselines to a cross-attention GNN–BERT fusion model and a contrastive dual-encoder. Cross-attention fusion (Task 3) achieved the strongest tagging performance (Macro-F1 0.611), improving over both unimodal baselines and a naive concatenation-based fusion, while confirming through ablation that language is the dominant signal when audio is represented only by chroma. Contrastive retrieval (Task 4) proved substantially harder at our compute scale, though it recovered a non-trivial zero-shot tagging signal. These results are consistent with the assignment’s own illustrative benchmark table and provide a reproducible, leakage-aware baseline for future extensions of this project.

References

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